

Precalc Lesson 1.7 Homework

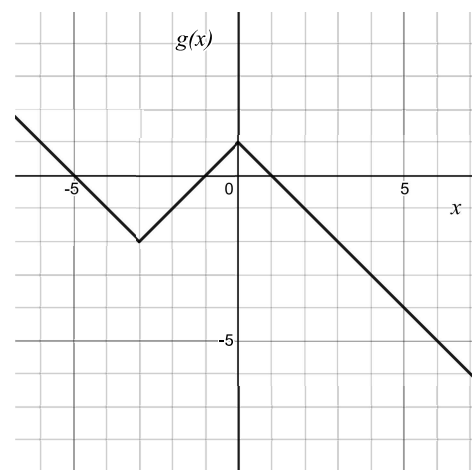
Name _____

1. Let $f(x) = x^2 + 3x - 1$ and $g(x) = 10 - 2x$. Find $f(x) - g(x)$.
2. Let $f(x) = \sqrt{20 - x}$ and $g(x) = \sqrt{10x - 4}$. Let $h(x) = f(x)g(x)$. Find $h(4)$.
3. Paul and Polly sell lemonade in front of their house in the summer. During the month of June, they sold about 26 cups of lemonade a day at a price of \$0.50 per cup. In July, they plan to increase their cups sold by 2 cups every day with some clever new advertising. They will also decrease the price per cup by 1 cent every day. Write an equation for a function $L(d)$ that gives the amount of money Paul and Polly make selling lemonade on day d in July.

4. The graph of a function g is shown on the xy -plane on the interval $[-7, 7]$.

- a. A table of selected values is given for a different function h , which has only one real zero.

x	$h(x)$
-5	-2
-2	7
0	3
2	0
4	-13
7	-20



Let $f(x) = -2g(x) + 3h(x)$. Find $f(4)$.

- b. Let $j(x) = g(x)h(x)$. How many x -intercepts does j have on the interval $[-7, 7]$?

- c. Let $k(x) = \frac{g(x)}{h(x)}$. Which values are restricted from the domain of k ? Explain.

5. Kara enlisted her friend Frannie to help shovel all the driveways in her grandma's condominium community. It takes Kara 40 minutes to shovel a driveway and it takes Frannie 45 minutes to shovel the driveway. Write an equation for a function, d , that gives the number of driveways Kara and Frannie can shovel by working together for x hours.

6. Let $f(x) = x^2 - 16$ and $g(x) = 2x - 9$.

a. What is the domain of f ? What is the domain of g ?

b. Find $\frac{g(x)}{f(x)}$.

c. What is the domain of $\frac{g(x)}{f(x)}$?

7. Let $h(x) = \sqrt{25 - x^2}$ and $j(x) = 7x - 1$.

a. What is the domain of $h(x)$?

b. What is the domain of $j(x)$?

c. Let $k(x) = h(x) + j(x)$. What is the domain of $k(x)$?

d. Let $f(x) = \frac{h(x)}{j(x)}$. What is the domain of $f(x)$?